ENERGY EFFICIENCY MODULE-HEAT EXCHANGERS

HEAT EXCHANGER PERFORMANCE ASSESSMENT

Heat Exchangers can deteriorate over time due to off-design operation, fouling, scaling, or corrosion.

This tool periodically assesses thermal duty, heat transfer coefficients (U), effectiveness (ϵ), and hydraulic pressure drops (ΔP) for both Shell & Tube and Plate Exchangers.

Metadata & Operational Configuration

Select Heat Exchanger Type:

Shell & Tube Heat Exchanger (STHE)

Design Data Inputs (from Test/Spec Reports)

Flow Configuration:

Counterflow

No. of Shell Side Passes (N)

1

No. of Tube Side Passes (M)

2

Total Design Heat Duty (kW)

10495.85

Overall Heat Transfer Coefficient U_d (Design/Service) (kW/m^2-K)

1.1500

Design Fouling Margin (%)

Hot Fluid Side Pressure Drop Design (bar)

0.10

Cold Fluid Side Pressure Drop Design (bar)

0.29

Hot Fluid Temp Range (°C)

74.20

Cold Fluid Temp Range (°C)

60.00

Corrected LMTD Design (°C)

73.60

15.0

Operation Data Inputs

Hot Fluid Flow Inlet (kg/hr)
18271.30

Hot Fluid Temp Inlet (°C)
164.20

Hot Fluid Pressure Inlet (bar g)
6.00

Cold Fluid Flow Inlet (kg/hr)
157438.00

Hot Fluid Temp Outlet (°C)
90.00

Hot Fluid Pressure Outlet (bar g)
5.90

Cold Fluid Temp Inlet (°C)
20.00

Cold Fluid Pressure Inlet (bar g)
16.00

Cold Fluid Temp Outlet (°C)
80.00

Cold Fluid Pressure Outlet (bar g)
15.70

Fluid Properties

Hot Fluid Specific Heat Capacity (kJ/kg-°C)
2.84700

Latent Heat of Condensation (kJ/kg)
2068.00000

Cold Fluid Specific Heat Capacity (kJ/kg-°C)
4.00000

Latent Heat of Vaporisation (kJ/kg)
0.00000

Heat Transfer Area (A) Calculation

How do you want to provide the Heat Transfer Area (A) m^2?

- Calculate A (Tube Geometry)
- Input A Manually

Enter Heat Transfer Area (A in m^2)
124.00

Final Calculated Heat Transfer Area (A): 124.0000 m²

How do you want to provide the Correction Factor (F)?

- (•) Calculate F
- ( ) Input F Manually

Assessment Results

	Parameters	Units	Operation Data	Design Data
0	Heat Exchanger Type	-	Shell & Tube Heat Exchanger (STHE)	Shell & Tube Heat Exchanger (STHE)
1	Duty (Q)	kW	10495.87	10495.85
2	Hot Fluid Pressure Drop ΔPh	bar	0.10	0.10
3	Cold Fluid Pressure Drop ΔPc	bar	0.30	0.29
4	Temp Range Hot Fluid	°C	74.20	74.20
5	Temp Range Cold Fluid	°C	60.00	60.00
6	Corrected LMTD	°C	65.98	73.60
7	Heat Transfer Coefficient U	kW/m²-K	1.283	1.150

Effectiveness (ε)

0.4161

Capacity Ratio (r)

0.0000

Fouling Resistance (Rd)

-0.0900 m²K/kW

AI Performance Diagnostic Summary

Generate AI Assessment Summary

Diagnostic Summary

- **Fouling Margin Assessment:** The operating overall heat-transfer coefficient ($U_{op} = 1.283\ kW\ m^{-2}\ K^{-1}$) exceeds the design value ($U_d = 1.150\ kW\ m^{-2}\ K^{-1}$) and is only 3 % below the clean-condition coefficient ($U_{clean} = 1.322\ kW\ m^{-2}\ K^{-1}$). This is well within the 15 % design fouling allowance, so the exchanger is performing within its fouling margin and no immediate cleaning is required.
- **Thermal Effectiveness & Capacity Assessment:** The measured effectiveness ($E = 0.4161$) is typical for a steam-condensing heater operating with a very large heat-capacity ratio. The capacity ratio $r = 0.0000$ reflects the condensing hot side's effectively infinite heat capacity ($C_{max} \to \infty$), placing the unit in an asymmetric flow regime ($r \ll 1$). The exchanger is therefore extracting about 42 % of the maximum possible heat, which is consistent with the NTU-limited design.
- **Hydraulic Comparison:** Hot-side pressure drop is unchanged ($\Delta P_h = 0.10$ bar vs design 0.10 bar). Cold-side pressure drop is marginally higher ($\Delta P_c = 0.30$ bar vs design 0.29 bar), a 3 % increase that is within normal operational variance and does not indicate channel blockage or excessive fouling.
- **Maintenance Recommendation:** Because the fouling margin remains intact and pressure drops are nominal, no corrective cleaning or tube-bundle removal is presently justified. Continue routine monitoring of U-value, ΔP, and LMTD trends; schedule a standard inspection (e.g., tube-rodding or visual check) during the next planned outage to verify mechanical integrity.

Brief Engineering Insight

The modest 3 % drop in U_{op} relative to the clean baseline, together with the reduced corrected LMTD (65.98 °C vs. 73.60 °C), reflects the expected thermal penalty of a condensing steam side that supplies a constant temperature heat source. The calculated fouling resistance ($Rd = -0.090\ m^2\ K\ kW^{-1}$) is negative, indicating that the apparent loss is dominated by the inherent NTU limitation rather than true deposit buildup. With $r = 0.0$, the hot side's infinite heat capacity forces the effectiveness to be governed solely by the number of transfer units; the observed $E \approx 0.416$ aligns with the design NTU range for this application. Consequently, the primary driver of the observed performance is the thermodynamic configuration (phase-change heat source)